AQA A Level Further Mathematics (7367)

Topic **B: Complex Numbers**

Sub topic **B7: Loci**

Topic Questions

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Question Paper

Topic	**B: Complex Numbers**
Sub topic	**B7: Loci**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ____________________

Date : ____________________

Time : ____________________

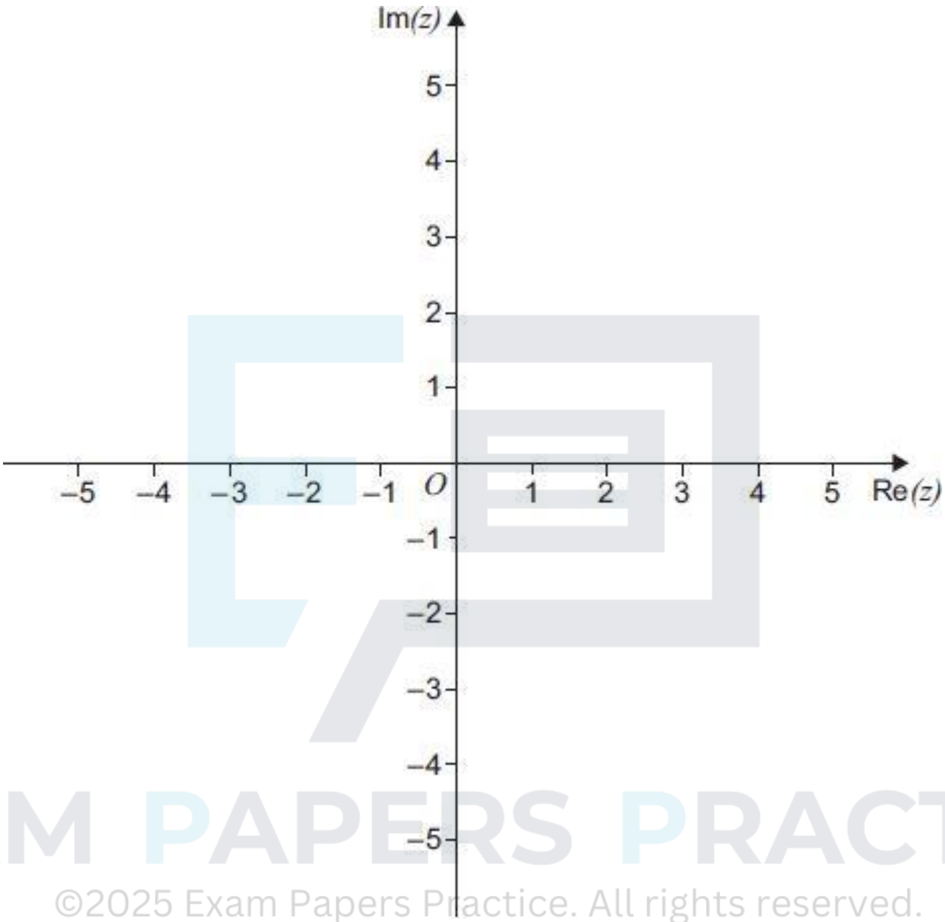
Total Marks Available : _____

Total Marks Received : _____

Q1.

(a) Sketch, on the Argand diagram below, the locus of points satisfying the equation

$$|z - 3| = 2$$



(1)

(b) There is a unique complex number w that satisfies both

$$|w - 3| = 2 \text{ and } \arg(w + 1) = \alpha$$

where α is a constant such that $0 < \alpha < \pi$

(i) Find the value of α.

(2)

(ii) Express w in the form $r(\cos\theta + \mathrm{i}\sin\theta)$.

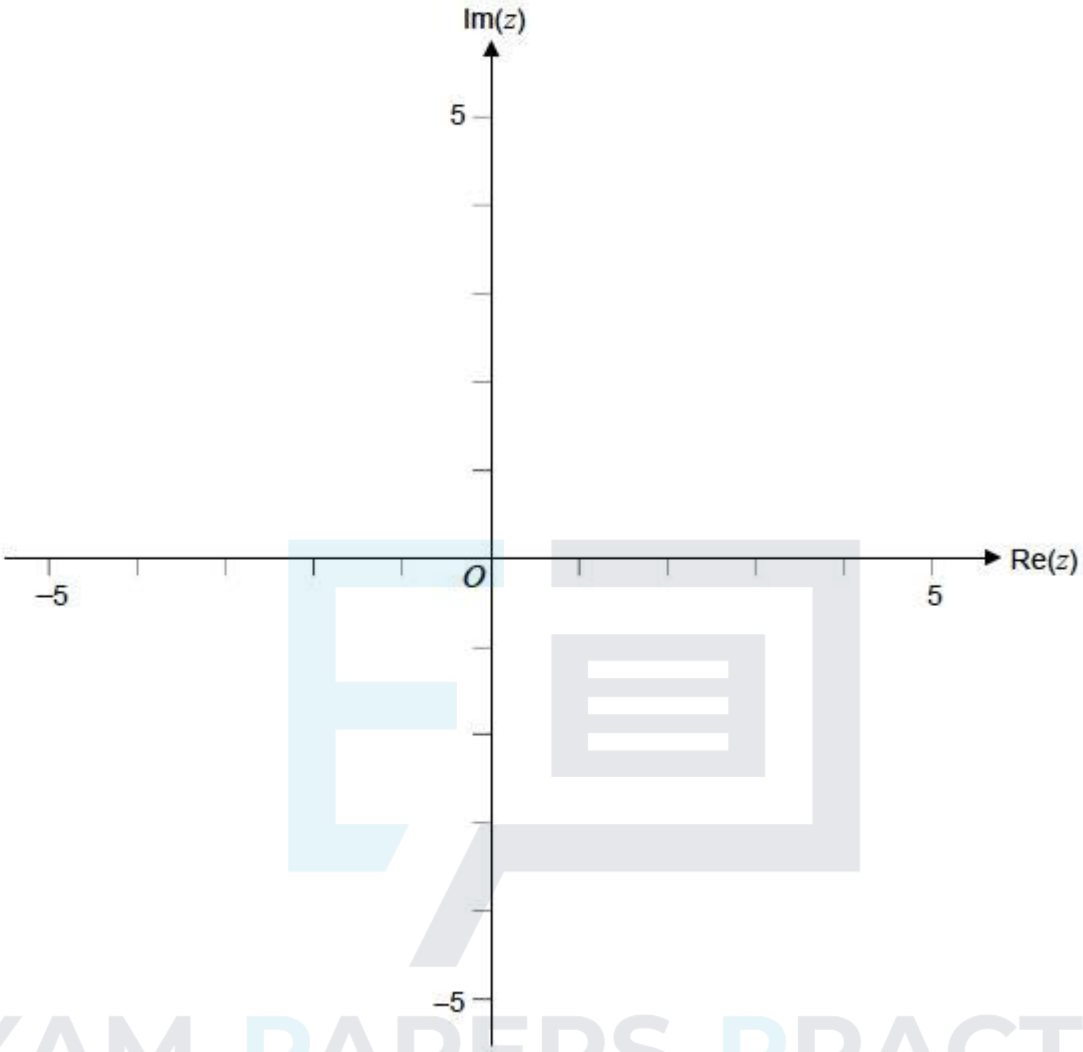
Give each of r and θ to two significant figures.

(4)

(Total 7 marks)

Q2.

(a) Sketch on the Argand diagram below, the locus of points satisfying the equation $|z - 2| = 2$



(2)

(b) Given that $|z - 2| = 2$ and $\arg(z - 2) = -\frac{\pi}{3}$, express z in the form $a + b\mathrm{i}$, where a and b are real numbers.

(3)

(Total 5 marks)

Q3.

Two loci, L_1 and L_2, in an Argand diagram are given by

$$L_1 : |z + 6 - 5\mathrm{i}| = 4\sqrt{2}$$

$$L_2 : \quad \arg(z + \mathrm{i}) = \frac{3\pi}{4}$$

The point P represents the complex number $-2 + \mathrm{i}$.

(a) Verify that the point P is a point of intersection of L_1 and L_2.

(2)

(b) Sketch L_1 and L_2 on one Argand diagram.

(6)

(c) The point Q is also a point of intersection of L_1 and L_2. Find the complex number that is represented by Q.

(2)
(Total 10 marks)

Q4.

(a) Sketch on an Argand diagram the locus of points satisfying the equation

$$|z - 6\text{i}| = 3$$

(3)

(b) It is given that z satisfies the equation $|z - 6\text{i}| = 3$.

(i) Write down the greatest possible value of $|z|$.

(1)

(ii) Find the greatest possible value of arg z, giving your answer in the form $p\pi$, where $-1 < p \leq 1$.

(3)
(Total 7 marks)

Q5.

(a) Draw on an Argand diagram the locus L of points satisfying the equation

arg $z = \frac{\pi}{6}$.

(1)

(b) (i) A circle C, of radius 6, has its centre lying on L and touches the line Re(z) = 0. Draw C on your Argand diagram from part (a).

(2)

(ii) Find the equation of C, giving your answer in the form $| z - z_0 | = k$.

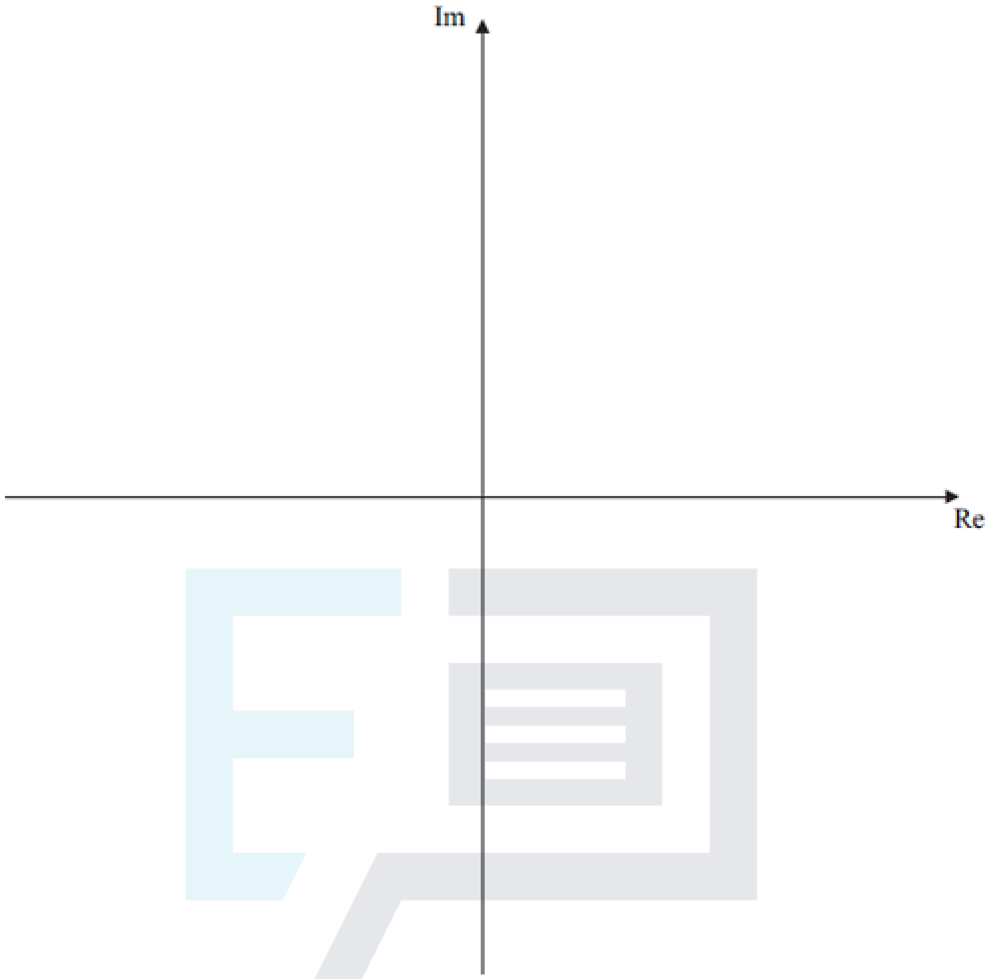
(3)

(iii) The complex number z_1 lies on C and is such that arg z_1 has its least possible value.
Find arg z_1, giving your answer in the form $p\pi$, where $-1 < p \leq 1$.

(2)
(Total 8 marks)

Q6.

(a) Draw on the Argand diagram below:



(i) the locus of points for which

$$|z - 2 - 3\mathrm{i}| = 2$$

(3)

(ii) the locus of points for which

$$|z + 2 - \mathrm{i}| = |z - 2|$$

(3)

(b) Indicate on your diagram the points satisfying both

$$|z - 2 - 3\mathrm{i}| = 2$$

$$\text{and} |z + 2 - \mathrm{i}| \le |z - 2|$$

(1)
(Total 7 marks)

Q7.

(a) Sketch on an Argand diagram the locus of points satisfying the equation

$$|z - 4 + 3\mathrm{i}| = 5$$

(3)

(b) (i) Indicate on your diagram the point P representing z_1, where both

$$|z_1 - 4 + 3i| = 5 \text{ and } \arg z_1 = 0$$

(1)

(ii) Find the value of $|z_1|$.

(1)
(Total 5 marks)

Q8.

(a) Draw on the same Argand diagram:

(i) the locus of points for which

$$|z - 2 - 5i| = 5$$

(3)

(ii) the locus of points for which

$$\arg(z + 2i) = \frac{\pi}{4}$$

(3)

(b) Indicate on your diagram the set of points satisfying both

$$|z - 2 - 5i| \leq 5$$

and

$$\arg(z + 2i) = \frac{\pi}{4}$$

(2)
(Total 8 marks)

Q9.

(a) On the same Argand diagram, draw:

(i) the locus of points satisfying $|z - 4 + 2i| = 4$;

(3)

(ii) the locus of points satisfying $|z| = |z - 2i|$.

(3)

(b) Indicate on your sketch the set of points satisfying both

$$|z - 4 + 2i| \leq 4$$

and

$$|z| \geq |z - 2i|$$

(2)
(Total 8 marks)

Q10.

Two loci, L_1 and L_2, in an Argand diagram are given by

$$L_1 : |z + 1 + 3i| = |z - 5 - 7i|$$

$$L_2 : \arg z = \frac{\pi}{4}$$

(a) Verify that the point represented by the complex number 2 + 2i is a point of intersection of L_1 and L_2.

(2)

(b) Sketch L_1 and L_2 on one Argand diagram.

(5)

(c) Shade on your Argand diagram the region satisfying

both $\quad |z + 1 + 3i| \le |z - 5 - 7i|$

and $\quad \frac{\pi}{4} \le \arg z \le \frac{\pi}{2}$

(2)
(Total 9 marks)

Q11.

(a) Indicate on an Argand diagram the region for which $|z - 4i| \le 2$.

(4)

(b) The complex number z satisfies $|z - 4i| \le 2$. Find the range of possible values of arg z.

(4)
(Total 8 marks)

Q12.

(a) Two points, A and B, on an Argand diagram are represented by the complex numbers 2 + 3i and –4 – 5i respectively. Given that the points A and B are at the ends of a diameter of a circle C_1, express the equation of C_1 in the form $|z - z_0| = k$.

(4)

(b) A second circle, C_2, is represented on the Argand diagram by the equation $|z - 5 + 4i| = 4$. Sketch on one Argand diagram both C_1 and C_2.

(3)

(c) The points representing the complex numbers z_1 and z_2 lie on C_1 and C_2 respectively and are such that $|z_1 - z_2|$ has its maximum value. Find this maximum value, giving your answer in the form $a + b\sqrt{5}$.

(5)
(Total 12 marks)

Q13.

A circle C and a half-line L have equations

$$|z - 2\sqrt{3} - i| = 4$$

and $$\arg(z + i) = \frac{\pi}{6}$$

respectively.

(a) Show that:

(i) the circle C passes through the point where $z = -i$;

(2)

(ii) the half-line L passes through the centre of C.

(3)

(b) On one Argand diagram, sketch C and L.

(4)

(c) Shade on your sketch the set of points satisfying both

$$|z - 2\sqrt{3} - \mathrm{I}\,| \leq 4$$

and $$0 \leq \arg(z + i) \leq \frac{\pi}{6}$$

(2)

(Total 11 marks)

Q14.

(a) A circle C in the Argand diagram has equation

$$|z + 5 - i| = \sqrt{2}$$

Write down its radius and the complex number representing its centre.

(2)

(b) A half-line L in the Argand diagram has equation

$$\arg(z + 2i) = \frac{3\pi}{4}$$

Show that $z_1 = -4 + 2i$ lies on L.

(2)

(c) (i) Show that $z_1 = -4 + 2i$ also lies on C.

(1)

(ii) Hence show that L touches C.

(3)

(iii) Sketch L and C on one Argand diagram.

(2)

(d) The complex number z_2 lies on C and is such that $\arg(z_2 + 2\text{i})$ has as great a value as possible.

Indicate the position of z_2 on your sketch.

(2)
(Total 12 marks)

Q15.

(a) Sketch on one diagram:

(i) the locus of points satisfying $|z - 4 + 2\text{i}| = 2$;

(3)

(ii) the locus of points satisfying $|z| = |z - 3 - 2\text{i}|$

(3)

(b) Shade on your sketch the region in which

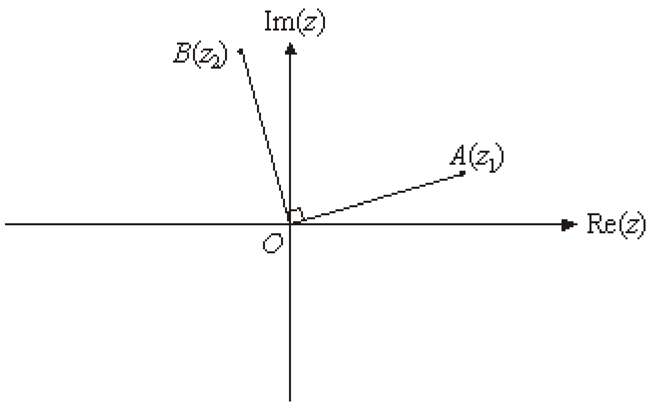
both $|z - 4 + 2\text{i}| \leq 2$

and $|z| \leq |z - 3 - 2\text{i}|$

(2)
(Total 8 marks)

Q16.

The sketch shows an Argand diagram. The points A and B represent the complex numbers z_1 and z_2 respectively. The angle $AOB = 90°$ and $OA = OB$.



(a) Explain why $z_2 = \text{i}z_1$.

(2)

(b) On a **single** copy of the diagram, draw:

(i) the locus L_1 of points satisfying $|z - z_2| = |z - z_1|$;

(2)

(ii) the locus L_2 of points satisfying $\arg(z - z_2) = \arg z_1$.

(3)

(c) Find, in terms of z_1, the complex number representing the point of intersection of L_1 and L_2.

(2)

(Total 9 marks)

Q17.

The complex number z satisfies the relation

$$| z + 4 - 4i | = 4$$

(a) Sketch, on an Argand diagram, the locus of z.

(3)

(b) Show that the greatest value of $| z |$ is $4(\sqrt{2} + 1)$.

(3)

(c) Find the value of z for which

$$\arg (z + 4 - 4i) = \frac{1}{6}\pi$$

Give your answer in the form $a + ib$.

(3)

(Total 9 marks)

Q18.

(a) On one Argand diagram, sketch the locus of points satisfying:

(i) $|z - 3 + 2i| = 4$;

(3)

(ii) $\arg(z - 1) = -\frac{1}{4}\pi$.

(3)

(b) Indicate on your sketch the set of points satisfying both

$$|z - 3 + 2i| \le 4$$

and $\arg(z - 1) = -\frac{1}{4}\pi$

(1)

(Total 7 marks)